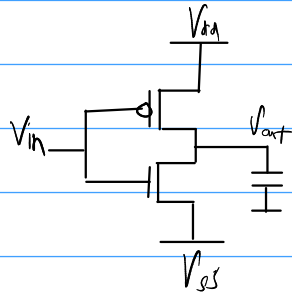


Circuit Design - CMOS Notes

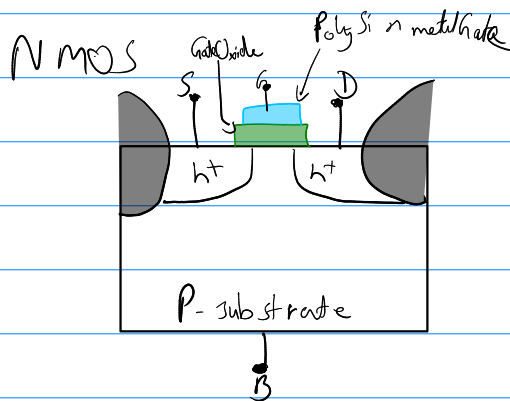


Input Slew is the rate at which an input signal changes with time / usually time taken by digital transformation $H[0] \rightarrow 1[0]$

$$\text{Input Slew} = \frac{\Delta V}{\Delta t}$$

$$\frac{V_{\text{out}} - V_{\text{in}}}{t_{\text{ao}} - t_{\text{io}}} = 0.008 \text{ V/ps}$$

Eg 0.1V to 0.9V in 10ps



- 4 terminal Device
- Consist of P-substrate
- Isolation Region (SiO_2)
- n^+ Diffusion Region
- Gate Oxide
- Poly Si or metal gate
- G - Gate, S - Source, D - Drain, B - Body

Threshold Voltage V_t

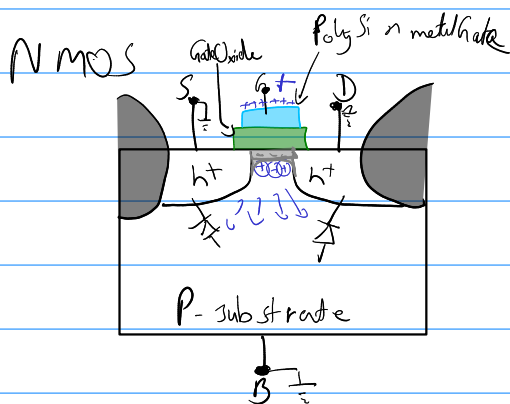
→ $V_{gs} = 0$

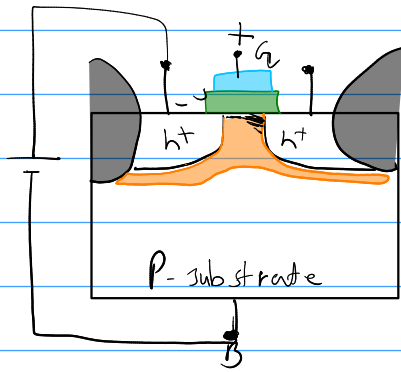
→ D, S, B to GND

→ Substrate - Source (B-S) and Substrate - Drain (B-D) form P-n junction diode

- Both junctions are 'off' due to 0V bias

- Hence, S-D resistance is 'high'





$V_{sb} = +ve \text{ value}$

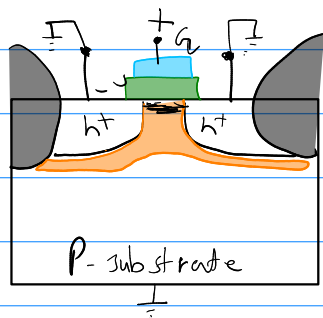
if additional voltage given (for good doping concentration)

$$V_{gs} = V_{t0} + \underline{V_t}$$

$$V_t = V_{t0} + \gamma \left(\sqrt{|-2\phi_f + V_{sb}|} - \sqrt{-2\phi_f} \right)$$

γ = body effect coefficient, expresses the impact of change in body bias V_{sb} (Unit in volts)

ϕ_f = Fermi potential

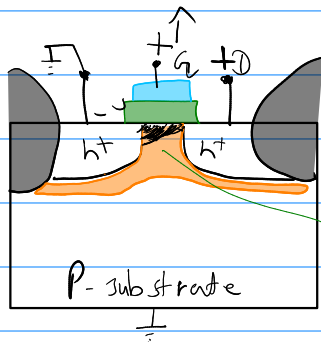


$V_{sb} = 0$

Resistive region of operation with small drain-source voltage

Pinch off region - no channel near drain region

Pinch off condition $V_{gs} - V_{ds} \leq V_t$



Voltage over the channel remains constant = $V_{gs} - V_t$

$$I_d = k_n/2 (V_{gs} - V_t)^2 \Rightarrow \text{linear f}^{\text{of}} V_{gs}$$

Drain current is constant no longer funcⁿ of V_{ds}

Effective conductive channel length is modulated by applied V_{ds}

$V_{ds} \uparrow$ Depletion region at drain $\uparrow$ Effective channel length $\downarrow$